

Survey Paper: AI-Powered Real-Time Urban Traffic Congestion Prediction Using Distributed Big Data Processing

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Abstract— Urban traffic congestion is a critical challenge affecting mobility, fuel efficiency, and environmental sustainability in rapidly growing cities. With the advent of big data and real-time data streams from sensors, GPS, and IoT devices, there is immense potential to develop intelligent traffic management systems. This survey explores the use of deep learning models for real-time traffic congestion prediction, focusing on architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), Graph Neural Networks (GNNs), and Transformers. Unlike traditional machine learning approaches, deep learning enables automatic feature extraction from complex spatiotemporal traffic data, leading to improved prediction accuracy and scalability. We also review the role of big data platforms like Apache Kafka and Apache Spark in supporting real-time ingestion and processing of high-velocity traffic data. This paper provides a comprehensive overview of recent advances, benchmark datasets, deployment frameworks, challenges, and future directions for implementing deep learning-driven, real-time traffic prediction systems in smart city infrastructures.

Keywords: *Traffic Congestion Prediction, Deep Learning, Spatiotemporal Modeling, Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN), Graph Neural Networks (GNN), Transformers, Big Data Analytics, Real-Time Data Streams, Apache Spark, Apache Kafka, Urban Traffic Management, Intelligent Transportation Systems (ITS), Time Series Forecasting, Smart Cities*

I. INTRODUCTION

Urbanization and rapid population growth have led to a surge in vehicular traffic, resulting in frequent congestion across metropolitan areas. Traffic congestion not only delays commute times but also increases fuel consumption, air pollution, and economic losses. Traditional traffic control systems, which rely on pre-defined rules or classical statistical models, often fail to adapt to the dynamic nature of real-world traffic scenarios.

Recent advancements in big data technologies and deep learning (DL) have opened new frontiers in building intelligent transportation systems (ITS). Massive volumes of real-time traffic data are now continuously generated by heterogeneous sources—such as inductive loop detectors, GPS-equipped vehicles, mobile devices, surveillance cameras, and Internet of Things (IoT) sensors. These data streams provide a rich foundation for predictive modeling, but their high velocity, variety, and volume pose significant challenges in terms of storage, processing, and real-time analysis.

To address this, big data frameworks like Apache Kafka (for real-time data ingestion) and Apache Spark (for distributed stream processing) are widely used to handle high-throughput, low-latency data pipelines. On top of these platforms, deep learning models such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), Graph Neural Networks (GNNs), and Transformers are increasingly being employed to learn complex, non-linear spatiotemporal patterns in traffic flows and predict future congestion states.

Compared to traditional machine learning, deep learning offers superior feature extraction, scalability, and adaptability—particularly when dealing with high-dimensional and unstructured traffic data like images, sensor logs, or GPS traces. For instance, CNNs can process traffic camera feeds, LSTMs can learn from sequential traffic flow data, and GNNs can model traffic networks as graphs to improve prediction accuracy at the road segment level.

This survey presents a comprehensive review of the current state-of-the-art in real-time traffic congestion prediction using deep learning and big data technologies. It examines the various deep learning architectures employed in this domain, the role of real-time data infrastructures, benchmark datasets, evaluation metrics, deployment strategies, and research gaps. The goal is to highlight how DL-driven approaches can contribute to building robust, scalable, and responsive traffic management solutions for future smart cities. With the advent of big data and artificial intelligence (AI), there has been a

paradigm shift in how urban traffic systems are monitored and optimized. The proliferation of real-time data sources—such as road sensors, GPS-enabled vehicles, mobile devices, and surveillance cameras—has opened new avenues for intelligent traffic prediction and control. However, the sheer volume, velocity, and variety of this data demand scalable and efficient processing frameworks.

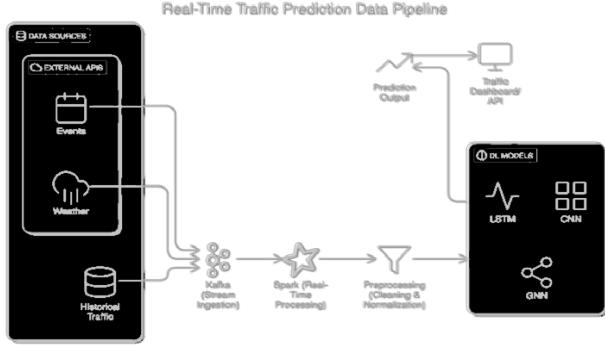


FIGURE 1. General Architecture for Real-Time Traffic Congestion Prediction Using Bigdata and Deep Learning

Big data platforms like Apache Kafka and Apache Spark have emerged as powerful tools for handling high-speed, real-time traffic data streams. Coupled with advanced machine learning (ML) and deep learning (DL) algorithms—such as Support Vector Machines (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Linear Regression, and Long Short-Term Memory (LSTM) networks—these systems can uncover complex spatio-temporal patterns in traffic flow and forecast future congestion with high accuracy.

This survey aims to provide a comprehensive overview of the current state-of-the-art approaches in real-time traffic congestion prediction using big data and AI. We examine the architectures, algorithms, technologies, and applications that underpin modern intelligent traffic systems. Furthermore, we identify key challenges, research gaps, and potential directions for future work in building scalable, real-time, and AI-enhanced traffic prediction systems suitable for smart city infrastructures.

II. LITERATURE SURVEY

Noureen Zafar et al. [1] proposes a hybrid GRU-LSTM based deep learning model and applied it on city-wide novel traffic data integrated from heterogeneous sources. It devises indigenous data pipeline that is composed of a set of algorithms dealing with map matching, sparsity handling, outlier removal, zero speed adjustments, Open Street Map (OSM) and segment mapping etc. Extensive experimentations have been carried out to demonstrate the improved performance of the proposed method. It describes a mechanism to integrate multiple sources of data into a hybrid feature space. Basically, it utilizes an ETA based congestion index as a road network state evaluation indicator that distributes the traffic state primarily into three categories ranging from smooth to congested to Highly Congested class. It integrates the traffic load, GPS, weather, special conditions

with the OSM data set and employed different deep learning and machine learning algorithms. Among classical learning techniques, Random Forest provided the best results whereas in deep learning algorithms, GRU-LSTM proved to be the best with the highest accuracy. Finally, it concluded that deep learning techniques are the most reliable learning techniques providing maximum accuracy and better yield than classical learning techniques when are applied in traffic congestion problems on arterial roads. These further paves way towards automatic labeling of the classes instead of using congestion indexes and automatic optimization of hyper parameters using adaptive techniques in future studies.

Jesús Mena-Orej et al. [2] conducted a comprehensive study comparing modern deep neural network architectures for traffic prediction. This work evaluates various advanced models under identical conditions using real-world traffic datasets, including scenarios of both normal flow and traffic congestion. The findings reveal that a model's performance in regular conditions does not necessarily reflect its effectiveness during emerging congestion scenarios. The study emphasizes the importance of assessing prediction models under congestion-specific circumstances. It also highlights the advantage of incorporating both spatial and temporal traffic dynamics over solely temporal analysis, especially for enhancing accuracy during congestion. Additionally, models that incorporate error feedback mechanisms, referred to as error recurrent models, consistently outperformed those that do not, regardless of the traffic condition.

Saurabh Pratap Singh Rathore et al. [3] proposes an AI-driven system designed to predict and manage traffic congestion. The system leverages continuous traffic data from IOT devices, such as images, GPS, and inductive loop sensors, to monitor real-time traffic conditions. Traffic predictions are made using a combination of CNN and LSTM networks. Additionally, Geographic Information System (GIS) technology is integrated for spatial analysis and dynamic traffic control, enabling real-time traffic light adjustments and providing drivers with alternative routes to avoid congestion. Preliminary experiments show that intelligent traffic management systems can significantly improve traffic flow and reduce congestion in large cities. The paper also explores challenges related to data privacy, scalability, and future advancements, including the integration of self-driving vehicles and the application of reinforcement learning. hence, the traffic congestion prediction and management system using Artificial intelligence provided a solution to the increased congestion in traffic systems in urban areas. By integrating IoT sensor data feeds that display real-world scenarios, the proposed system can employ deeper and more advanced constructs such as CNN LSTMs for further and more precise traffic pattern prediction and, consequently, a more dynamic traffic flow control. The integration of GIS is also applied to improve spatial analysis and the ability to make improved routing decisions. At the same time, real-time data processing enables the system to adapt to signal timings and rerouting changes quickly.

Malathi et al. [4] presents an innovative approach to managing traffic congestion through an intelligent system that combines

advanced congestion detection with adaptive signal control. The system leverages artificial intelligence to process various data sources, including real-time traffic flow, road conditions, and historical data, enabling it to detect congestion patterns with high accuracy. Machine learning algorithms are employed to analyze these patterns, allowing the system to automatically assess congestion levels and optimize signal timing dynamically based on factors such as traffic volume, queue length, and delay. The effectiveness of this approach has been validated through simulation studies, where the system demonstrated impressive performance metrics: an accuracy of 94.89%, a precision of 97.40%, a recall of 96.38%, and an F1 score of 96.76%. These results highlight the system's capability to effectively manage traffic conditions by accurately detecting congestion and minimizing false positives. By integrating these advanced techniques, the proposed solution significantly enhances traffic flow, reduces travel time, and improves the overall driving experience. The Intelligent Traffic Congestion Identification and Signal Switching System presents a groundbreaking solution to the growing challenge of traffic congestion. By integrating advanced congestion detection techniques with adaptive signal control, this system offers a novel approach to mitigating traffic issues. The utilization of artificial intelligence enables real-time analysis of traffic patterns, allowing for the accurate identification of congested areas, which is essential for efficiently adjusting signal timing to optimize traffic flow. The system's adaptability and automation make it an effective and practical tool for managing congestion, thereby improving overall transportation efficiency and reducing travel times for commuters.

Hongsuk Yi et al. [5] developed an automated machine learning framework aimed at optimizing hyperparameters for traffic prediction models, particularly focusing on major highway systems. Recognizing the challenges associated with training deep learning models-such as the time and expertise required for hyperparameter tuning-they introduced the Hypernet framework. This framework leverages advanced techniques like Bayesian optimization and meta-learning to streamline the hyperparameter search process. By integrating Hypernet with Long Short-Term Memory (LSTM) networks, the study effectively captures temporal variations in traffic data, enhancing the prediction of highway congestion. The primary objective is to automate the training process, reducing reliance on manual configuration and improving model performance across diverse traffic datasets.

Humayun Kabir et al. [6] proposed an Intelligent Traffic Management System (ITMS) utilizing the Internet of Things (IoT) to tackle traffic congestion, a significant hurdle for modern cities. The ITMS gathers real-time traffic data (vehicle count, speed, lane occupancy) through strategically positioned sensors at intersections. A central control system analyzes this data and dynamically adjusts traffic signals, allocating green time based on actual traffic flow. This approach aims to optimize traffic flow, significantly reduce congestion, and improve efficiency. The paper explores the potential benefits of the proposed system, including shorter travel times, improved air quality, and enhanced safety for pedestrians and vehicles. The study proposes an intelligent technique for regulating and improving traffic signal systems that react in

real time to automobile traffic. Preventing congestion and ensuring traffic flow requires controlling intersections in densely populated regions. For a variety of signal cycles, it evaluates both the fixed and improved traffic lights.

Mouna Zouari Mehd et al. [7] proposes a differential-entropy-based approach, it is a supervised congestion prediction from traffic meta-parameters based on a convolutional neural network was proposed. Traffic parameters include node localization, date, day of the week, time of day, special road conditions, and holidays. The proposed model is validated on the City Pulse dataset, which is a set of vehicle traffic records. Simulation results on the City Pulse dataset illustrate that the proposed approach yields accurate prediction rates for different nodes considered. The proposed system can prevent traffic congestion by reorienting the drivers to follow other itineraries. The simulation results illustrate that the proposed approach yields accurate prediction rates for the different nodes considered.

Kavitha et al. [8] research introduces an IoT-based traffic control system designed to alleviate this issue by using RFID technology and Arduino microcontrollers. The system detects emergency vehicles via RFID tags and dynamically adjusts traffic signals to provide an uninterrupted path. Implemented with RFID receivers and Arduino controllers, the system ensures real-time management of traffic signals. Results from preliminary tests indicate a substantial reduction in transit times for emergency vehicles. This study demonstrates the potential of IoT-based solutions in improving urban traffic management, enhancing both public safety and emergency response efficiency. The integration of RFID technology and Arduino controllers enables real-time dynamic adjustments to traffic signals. While the system demonstrates considerable improvements, it has certain limitations. Potential interference issues from other RFID equipped vehicles and the dependency on the reliability of the RFID and Arduino components need to be addressed.

Yuan-yuan Chen et al. [9] discussed online open data to provide traffic related information. Traffic conditions collected from web-based map services are used to demonstrate the feasibility. The stacked long short-term memory model, a kind of deep architecture, is used to learn and predict the patterns of traffic conditions. Experimental results show that the proposed model for traffic condition prediction has superior performance over multilayer perceptron model, decision tree model and support vector machine model. Online open data are potential sources to be used to provide traffic related information. This paper reviews four categories of open data platforms, i.e., web map, social media, local events and weather website. Traffic condition data collected from web map provider to learn the hidden patterns of traffic conditions. It proposes a deep learning approach with a stacked LSTM model for traffic condition

prediction. Online training was applied during the test phrase. It was compared with performance of the proposed stacked LSTM model with MLP model, decision tree model and SVM model. Experiments show the proposed method is superior to the competing methods.

Leixiao Li et al. [10] proposed the MF-TCPV, it uses Data- X and DataV to implement the integration of multi-source heterogeneous traffic data and the visualization of congestion prediction results. A deep prediction model named LSTM-SPRVM based on deep learning algorithms, machine learning algorithms, and Spark parallelization technology for the prediction of traffic congestion features in the future is proposed. In MF-TCPV, traffic congestion is divided into six levels based on Fuzzy Comprehensive Evaluation and traffic congestion features such as average speed, road occupancy rate, and traffic flow density. The experimental results demonstrate that MF-TCPV is capable of predicting the traffic congestion accurately and displaying prediction results visually. LSTM-SPRVM is better than other existing deep learning models in terms of prediction accuracy, and MF-TCPV can intuitively visualize the prediction results of traffic congestion.

Naveen et al. [11] propose a model that has for predicting and visually representing congestion of traffic using machine learning as well as deep learning. This initiative with the purpose of examining traffic data to obtain insights for traffic regulation and planning. The process initiates with an initial exploration of data through an Exploratory Data Analysis (EDA) and to perceive the distribution and patterns within the dataset. Descriptive statistics, correlation analysis, and visual aids are employed for this purpose. Following the EDA phase, tactics for manifold learning are utilized to simplify the dataset while retaining crucial information. Identifying significant features influencing the prediction of traffic volume is facilitated by methods for feature importance and selection. The obtained features are rigorously evaluated to comprehend traffic patterns and traffic jam levels. Visual representations capture the distribution of traffic volume based on factors like time, day, and month, assisting in pinpointing congestion hotspots and optimizing traffic flow. This project underscores the powerful impact of data analysis and feature extraction in offering practical insights for traffic regulation strategies. Route Rover, with its machine learning driven approach to transportation routing, offers potential benefits such as enhanced travel efficiency and reduced congestion. Yet, its success hinges on the fidelity of real-time data and scalability considerations. With ongoing refinement, Route Rover could become a transformative tool for optimizing travel routes and mitigating traffic challenges.

Supriya Tel sang et al. [12] explores a novel method to traffic control that enhances signal control systems in metropolitan environments by applying machine learning techniques. This

research revolves around a methodology that improves vehicle flow and reduces traffic congestion. This paper elaborates on the methodology utilized for collection of data and vehicle detection methods used for optimizing traffic signals. It incorporates machine learning algorithms into traffic signal control systems offers a promising solution for enhancing urban traffic management. This model adapts signal timings dynamically, resulting in substantial improvements in overall traffic efficiency. Additionally, the system's ability to prioritize emergency vehicles highlights its potential to improve public safety and response times during critical situations. The successful implementation of our algorithm emphasizes the significance of innovative traffic management and the revolutionary effects of machine learning technology on the development of future metropolitan planning and infrastructure development.

Chun Yao Zhong et al. [13] proposes a congestion warning framework (based on the "data + model" framework), which is based on the fact that during the duration before and after congestion occurs, traffic increases sharply and speed decreases sharply. In order to find a stable and reliable warning point, this article first determines the range of speed and flow values before and after congestion by observing actual traffic flow data. At the same time, through data fitting, select an appropriate traffic flow model to describe the basic relationship between the traffic parameters of the selected road section. Then, a congestion warning model based on the slope change of the basic graph was proposed. Finally, within the preliminary warning range determined by historical data, the proposed model is used to identify early warning points and achieve congestion warning.

Suthaharan Satkunarajah et al. [14] propose a model that additionally utilizes an approach to dynamically adjust traffic light schedules in four-way intersections in real-time, taking into consideration variations in traffic density. It considers mainly two traffic scenarios, such as light traffic conditions and heavy traffic conditions. In this simulation experiment, it is clearly seen that our proposed approach of dynamically changing the traffic light schedule outperforms conventional traffic control with a fixed time duration. It Proposes a model with an algorithm that utilizes a simulation-based approach to automatically adjust traffic light schedules in real-time, considering variations in traffic density in each road. It examines the traditional Green signaling method, exploring its implementation with different total time duration of 80 Seconds, 60 Seconds, and 40 Seconds. It could be able to achieve the significant advantage when the highway traffic system is implemented using our proposed Green signaling method, which operates with a total cycle duration of 40 Seconds whereas the traditional Green signaling method performed lower with the fixed time duration of 20 Seconds per road. As a result, this approach would definitely reduce

traffic delays and congestion in the transportation management system.

Tianrui Zhang et al. [15] proposes a traffic state prediction method based on improved particle swarm optimization (IPSO) optimized radial basis function (RBF) and long- and short-term memory network (LSTM)/ support vector machine (SVM) feature fusion model was proposed for urban traffic state prediction. It verifies the superiority of the feature fusion model based on IPSO-RBF and LSTM/SVM in this paper. And also it was aiming at the problem of urban traffic congestion, a congestion section control method based on traffic allocation is proposed. Through VISSIM simulation and comparison with other traffic control schemes, the superiority of the congestion section optimization method proposed. It studies traffic state prediction and congestion control by combining the theory of traffic prediction and control and the theory of intelligent algorithm. The factors that affect the traffic state are determined by the use of traffic related theory knowledge. It aiming at the problem of traffic state prediction, a traffic state prediction model based on IPSO-RBF and LSTM/SVM feature fusion is proposed. And also the traffic congestion problem, a traffic congestion control method based on traffic allocation is proposed.

Binghua Nie et al. [16] discusses a traffic flow control simulation model and designs a fuzzy control algorithm to simulate the traffic flow state on highways. Subsequently, a comparative analysis is conducted on the traffic states under different strategies, including no-control, ALINEA control, and fuzzy control. To further demonstrate the effectiveness of fuzzy control, it relies on real data from the Xibao Expressway to simulate the traffic flow under fuzzy control. The results indicate that under equivalent conditions, the fuzzy control method can achieve superior performance. It has developed a series of control strategies for regulating traffic flow to improve the operational state of traffic. Using real data from the Xibao Highway as a foundation, simulations were conducted to assess traffic flow under fuzzy control. The simulation results demonstrate that fuzzy control effectively enhances the operational state of traffic flow and plays a role in mitigating congestion on the highway. In comparison to the traditional ALINEA control algorithm, fuzzy control exhibits a smoother, more stable effect, coupled with a faster response rate. However, it is worth noting that the rule formulation in the fuzzy control algorithm is easily defined, modified, or deleted. Nevertheless, its formulation relies heavily on human experience, posing the potential challenge of insufficient or redundant rules.

Shyam Shankaran et al. [17] proposes a Adaptive Traffic Control System (ATCS), it is a traffic management technique that modifies or adapts the timing of traffic signals based on the real demand for traffic and achieved using a control system that includes both hardware and software, where hardware is the sensor used for real-time traffic density estimation and software is designed using captured data analysis of the city's current traffic flow. It depicts a model of camera-based traffic monitoring and processing system which reduces the cycle time and possesses special provisions for emergency vehicles. The key benefit of the ATCS framework is that it is stable and easy to install. Using image processing technique, the calculation of the length of the queue across junctions can prove to be more accurate than conventional inferential sensing devices and infrared sensors. The ATCS system also help to reduce the level of noise and air pollution, resulting in low fuel consumption as well. This system shows that it can stabilize traffic in 4-way intersection squares in far less cycles than in traditional fixed time timers with less energy and maintenance. There is an immense number of developments in the modern form of an adaptive traffic control system which might include AI based image processing. The system can be significantly enhanced with the introduction of techniques such as solar operated devices and sensors, LED display boards at junctions, synchronization of all traffic signals in the city and high- resolution cameras with enhanced night vision capabilities.

Yu Zheng et al. [18] explored diverse data sources in Intelligent Transportation Systems (ITS) and proposed a scalable big data analysis framework using distributed computing and parallel processing. The system efficiently processes large-scale traffic data in real time, offering critical decision support for intelligent traffic control. Experimental evaluations demonstrated the framework's scalability and effectiveness in data preprocessing. The study emphasizes that accurate traffic flow prediction enhances signal timing, congestion management, and road efficiency. However, challenges such as handling the increasing scale and complexity of data remain, highlighting the need for continued research in scalable big data processing.

TABLE 1. Comparison table for literature Review

Reference	Methodology/Model	Data Sources/Features	Key Contributions	Limitations
[1]	Hybrid GRU-LSTM, Random Forest	GPS, weather, OSM, ETA index, traffic load, conditions	GRU-LSTM outperformed classical models; handles sparse data well	Not suitable for non-linear non-sequential data; data sparsity still a challenge
[2]	DNNs vs Error Recurrent Models	Real traffic data under varied conditions	Error recurrent models outperform DNNs; spatiotemporal modeling improves accuracy	Doesn't address edge deployment or real-time latency
[3]	CNN + LSTM, GIS + Reinforcement Learning (future)	IoT sensor data (GPS, images, loop sensors), GIS	Real-time traffic rerouting using GIS; integration of CNN+LSTM for spatial-temporal prediction	Future scope (RL, GIS) is not implemented; hardware integration challenges
[4]	ML classification + Adaptive Signal Control	Traffic flow, queue length, past congestion	Achieved 94.89% accuracy; optimizes traffic flow via adaptive signals	Dataset limited to one location; lacks real-time responsiveness
[5]	AutoML + HyperNet + Bayesian Optimization	Highway traffic datasets	Auto-optimized LSTM configurations; reduces training effort	Requires significant computational resources; less interpretable
[6]	IoT-based Intelligent Traffic System	Sensor data: vehicle count, speed, lane occupancy	Real-time dynamic signal control; improves air quality and safety	Requires reliable sensor networks; not ML-based
[7]	CNN + Differential Entropy	CityPulse dataset: date/time, location, road status	Effective supervised prediction; driver rerouting suggestion feature	Limited to static dataset; may not adapt to unseen conditions
[8]	IoT with RFID and Arduino	RFID sensors for emergency vehicles	Real-time priority for ambulances; simple hardware integration	Interference issues with RFID; no predictive component
[9]	Stacked LSTM + Online Learning	Web maps, weather, events, social media	Dynamic learning on test data; performs better than MLP, DT, SVM	Lacks deployment details; online learning increases model complexity
[10]	LSTM-SPRVM + MF-TCPV Platform	Data-X, DataV, Spark; heterogeneous traffic data	Multi-level classification; high accuracy and visualization	Requires advanced infrastructure (Spark, DataV); limited scalability for small cities
[11]	ML + DL with EDA	Time, day, month, visual EDA	Strong visual and statistical analysis; model integration	No real-time testing; lacks comparison with DL models
[12]	ML-enhanced Dynamic Signal Timing	Live traffic and vehicle detection	Real-time emergency handling; efficient signal control	Not scalable; lacks cloud or edge deployment strategy

[13]	Data-driven Congestion Warning Framework	Speed-flow pre- and post-congestion	Early congestion detection via speed-flow shift modeling	Only works post-congestion; not proactive
[14]	Video Surveillance + Sensor-based Prioritization	CCTV footage, ultrasonic + IR sensors	Emergency vehicle detection and signal override	Requires costly camera infrastructure; lacks predictive ML models
[15]	YOLOv3 + LSTM + YOLOv5 for congestion severity	CCTV traffic images	Visual classification of congestion levels; ML-based severity detection	Real-time latency and large GPU requirements
[16]	Wireless Sensors + Dynamic Signal Control	Wireless node data	Real-time green light optimization; adaptive based on vehicle flow	Sensor nodes prone to failure; limited area coverage
[17]	ANN, SVM, KNN Comparison	Speed, travel time, vehicle count	ANN outperformed SVM and KNN; good for small-scale prediction	Dataset not disclosed; limited scalability
[18]	IoT Sensors + Traffic Priority System	Sound sensors, vibration sensors, IR	Priority for emergency vehicles and adaptive signaling	No ML/AI usage; less accurate compared to DL methods

III. PROPOSED METHODOLOGY

The objective of the proposed system is to design a robust framework for real-time traffic congestion prediction using various deep learning architectures. Unlike conventional methods that rely on static datasets, this approach integrates API-based live traffic data acquisition and dynamic model evaluation. The methodology consists of five key stages: Data Acquisition, Preprocessing, Feature Engineering, Model Design & Training, and Evaluation & Deployment.

A. Data Acquisition:

Traffic-related data is collected in real-time from publicly available APIs such as TomTom, HERE Maps, Google Maps API or OpenTraffic. These APIs provide dynamic information including:

- Current speed S_c
- Free flow speed S_f
- Travel time T
- Geographic coordinates
- Jam factor or traffic flow indicators (if available)

In cases where a congestion index C_i is not provided, it is computed using the formula:

$$C_i = 1 - \frac{S_c}{S_f}$$

Where:

- S_c = current observed speed
- S_f = free-flow or Ideal speed
- $C_i \in [0,1]$, with higher values indicating greater congestion

B. Data Preprocessing

The acquired data is structured and cleaned to remove noise and inconsistencies. The following steps are applied:

1. Missing Value Handling: Forward fill or linear interpolation is used.
2. Normalization: Features are scaled using Min-Max normalization:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

3. Timestamp Alignment: Timestamps are converted to a uniform time zone and format.
4. Sequence Framing: A sliding time window approach is used to create sequences for temporal prediction. For a window of size w :

$$X_t = \{x_{t-w+1}, \dots, x_t\}, \quad y_t = C_{i,t+1}$$

C. Feature Engineering

Along with primary features, derivative features such as speed variability, hour of the day, weekday/weekend flag, and weather data (optional) are considered to enrich model input.

$$\text{Speed Variability} = \frac{\sigma(S_c)}{\mu(S_c)}$$

Where:

- $\sigma(S_c)$ is the standard deviation of current speed
 - $\mu(S_c)$ is the mean speed over the window
- D. Deep Learning Models Explored

Multiple deep learning models are evaluated to determine the most effective architecture for predicting congestion in real time. These include:

1. Long Short-Term Memory (LSTM)
2. Gated Recurrent Unit (GRU)
3. Convolutional Neural Networks (CNN) for time-series
4. Transformer-based models (e.g., Temporal Fusion Transformer)
5. Hybrid Models (e.g., CNN-LSTM)

Each model is trained using the processed time-series data to predict either:

- A continuous congestion index
- Or a congestion class (e.g., low, medium, high)

The training loss function depends on the output type:

- For Regression

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

- For classification

$$L_{CrossEntropy} = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

E. Model Evaluation and Deployment

Models are evaluated using performance metrics like:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Classification Accuracy (if applicable)
- F1-Score
- Inference time (for real-time feasibility)

The best-performing model is deployed in a real-time prediction pipeline. Live sequences are passed to the model, and outputs are visualized via:

- Traffic heatmaps

- Time-series charts

IV. SUMMARY

The project focuses on the development of a real-time traffic congestion prediction system using deep learning techniques, leveraging dynamic traffic data obtained through public APIs. The primary objective is to accurately predict traffic congestion levels in urban areas and provide timely insights for effective traffic management and route optimization.

To achieve this, the proposed framework incorporates a comprehensive pipeline comprising real-time data acquisition, preprocessing, feature engineering, model training and evaluation, and deployment. Traffic data, including current speed, free-flow speed, and travel time, is continuously fetched from sources like the TomTom API, which forms the foundation of the dataset. The congestion index is computed and used as the target variable for prediction.

A variety of deep learning models—including LSTM, GRU, CNN, and Transformer-based architectures—are explored to assess their performance in modelling temporal traffic patterns. The system evaluates both regression and classification approaches, depending on whether the output is a continuous congestion index or a categorical congestion level.

Extensive experimentation is conducted to identify the model that offers the best trade-off between prediction accuracy and real-time inference speed. The final output is visualized through an intuitive dashboard displaying congestion alerts, traffic heatmaps, and predictive analytics.

This system has significant real-world applications in smart city development, urban traffic monitoring, emergency response planning, and navigation services, thereby contributing to reduced travel time, improved safety, and enhanced commuter experience.

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